R1

Pierson Lipschultz

March 23, 2026

1 New Code / Devs

1.1 Python Terminal Script for Filtering

Because we're likely to need to filter quite a few systems with very similar methods I decided to make an actual script instead of just using a notebook. The largest upside is that this can easily be run on a compute node, which should make filtering larger sims easier.

Takes a user input in the CLI (see below), filter for all `BH_Sol` systems, and returns the population file with said system as well as a .csv of just the `BH_Sol` rows. Furthermore, as it is running it provides an output via terminal stating how many `BH_MS` systems it found, how many are S1:S2, as if any are S2:S1, it prints a warning as well as listing the count of them.

Useage Example

```
python BH-Sol_Filter.py /Data/1e+00_Zsun_population.h5 -o ./testResults
```

Arguments

Argument	Use	Ex value
`-o`	output location	.results
`-outputName`	name for the outputed files[1]	`BH_Sol_Systems`
`-overwrite`	Whether to overwrite existing results, T/F bool	True
`--maxMass`	Max mass used when filtering, defaults to $3M_\odot$	3.5
`--maxPeriod`	Max period used when filtering, defaults to 3.5 days	4

Can be found at https://github.com/PiersonLip/stellarBHs/blob/main/FilteringScripts/BH-Sol_Filter.py

[1] If a name is not specificed, it will use the orignal filename, and append a specifier to the front

2 New Research

When filtering via notebook and via sciript, both returned 6 systems that had $p_{orb} < 3.5, M_{\odot,2} < 3, t < 10e9$

These systems all evolved via the same channel, that being `ZAMS_oCE1_CC1_oRLO2_CC2`. Via inspection of the history df and the oneline, it is evident that the systems evolve detached, then post-MS **unstable** RLO happens via S1 $\rightarrow$ S2. This leads to common envolope, which *greatly* decreases the orbtial period. This envolope is then ejected, leaving a depleted stripped He core. The core then collapses in a couple thousand years, leaving the system in a BH-MS state with low p_{orb}/seperation.

These systems then evolve in a semi-partiuclar way, where it is highy likely that there will be RLO from S2 onto S1, and because S1 ¿ S2, this is stable, and leads to *very* prolonged stable RLO (seemingly on the scale of $> 14Gyr$), which leaves S2 at low masses ($< .08M_{\odot}$).

Interestly, the inital period for all the systems is very close to $4000d$, with mean of 4022 and a std of only 151. However, a larger sample size is needed to confirm this.

Additionally, the inital masses of S1 and S2 appear to be quite constrained as well.

	orbital_period_i
count	6.000000
mean	4022.716197
std	151.522384
min	3873.118288
25%	3908.300983
50%	3978.305326
75%	4124.397101
max	4248.507680

	S1_mass_i
count	6.000000
mean	17.515761
std	0.520269
min	16.815111
25%	17.190036
50%	17.510176
75%	17.818155
max	18.254965

	S2_mass_i
count	6.000000
mean	2.296788
std	0.320702
min	1.797809
25%	2.120753
50%	2.353828
75%	2.547210
max	2.620529